

emotion-detection-2 (1)

April 4, 2025

```
[1]: !pip install ultralytics  
      !pip install opencv-python  
      !pip install torchvision  
      !pip install torch
```

Collecting ultralytics

Downloading ultralytics-8.3.101-py3-none-any.whl.metadata (37 kB)

Requirement already satisfied: numpy<=2.1.1,>=1.23.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (1.26.4)

Requirement already satisfied: matplotlib>=3.3.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (3.7.5)

Requirement already satisfied: opencv-python>=4.6.0 in
/usr/local/lib/python3.10/dist-packages (from ultralytics) (4.10.0.84)

Requirement already satisfied: pillow>=7.1.2 in /usr/local/lib/python3.10/dist-
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Requirement already satisfied: torchvision>=0.9.0 in
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Requirement already satisfied: tqdm>=4.64.0 in /usr/local/lib/python3.10/dist-
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/usr/local/lib/python3.10/dist-packages (from ultralytics) (0.12.2)

Collecting ultralytics-thop>=2.0.0 (from ultralytics)

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Requirement already satisfied: contourpy>=1.0.1 in

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Downloading ultralytics_thop-2.0.14-py3-none-any.whl (26 kB)

Installing collected packages: ultralytics-thop, ultralytics

Successfully installed ultralytics-8.3.101 ultralytics-thop-2.0.14

Requirement already satisfied: opencv-python in /usr/local/lib/python3.10/dist-packages (4.10.0.84)

Requirement already satisfied: numpy>=1.21.2 in /usr/local/lib/python3.10/dist-packages (from opencv-python) (1.26.4)

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 Requirement already satisfied: MarkupSafe>=2.0 in
 /usr/local/lib/python3.10/dist-packages (from jinja2->torch) (3.0.2)

```
[4]: import os
      print(os.listdir("/kaggle/input/")) # List available datasets
```

```
['yolov9s.pt', 'fer2013']
```

```

[5]: import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
import matplotlib.pyplot as plt
import os

# Dataset path in Kaggle (update if needed)
dataset_path = "/kaggle/input/fer2013/train"

# Image size and batch size
img_size = (64, 64)
batch_size = 32

# Load dataset
train_ds = keras.preprocessing.image_dataset_from_directory(
    dataset_path,
    validation_split=0.2,
    subset="training",
    seed=42,
    image_size=img_size,
    batch_size=batch_size
)

val_ds = keras.preprocessing.image_dataset_from_directory(
    dataset_path,
    validation_split=0.2,
    subset="validation",
    seed=42,
    image_size=img_size,
    batch_size=batch_size
)

# Normalize pixel values (0-255 -> 0-1)
normalization_layer = layers.Rescaling(1./255)

train_ds = train_ds.map(lambda x, y: (normalization_layer(x), y))
val_ds = val_ds.map(lambda x, y: (normalization_layer(x), y))

# Build CNN Model
model = keras.Sequential([
    layers.Conv2D(32, (3, 3), activation="relu", input_shape=(64, 64, 3)),
    layers.MaxPooling2D(),

    layers.Conv2D(64, (3, 3), activation="relu"),
    layers.MaxPooling2D(),

    layers.Conv2D(128, (3, 3), activation="relu"),

```

```

        layers.MaxPooling2D(),

        layers.Flatten(),
        layers.Dense(128, activation="relu"),
        layers.Dropout(0.5),
        layers.Dense(len(train_ds.class_names), activation="softmax") # Output_
↪layer
    ])

# Compile the model
model.compile(optimizer="adam",
              loss="sparse_categorical_crossentropy",
              metrics=["accuracy"])

# Train the model
history = model.fit(train_ds, validation_data=val_ds, epochs=10)

# Save the trained model
model.save("cnn_emotion_model.h5")

# Plot training history
plt.plot(history.history['accuracy'], label="Training Accuracy")
plt.plot(history.history['val_accuracy'], label="Validation Accuracy")
plt.legend()
plt.show()

```

Found 28709 files belonging to 7 classes.

Using 22968 files for training.

Found 28709 files belonging to 7 classes.

Using 5741 files for validation.

/usr/local/lib/python3.10/dist-

packages/keras/src/layers/convolutional/base_conv.py:107: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

super().__init__(activity_regularizer=activity_regularizer, **kwargs)

```

-----
AttributeError                                Traceback (most recent call last)
<ipython-input-5-6310aee7123b> in <cell line: 40>()
     51     layers.Dense(128, activation="relu"),
     52     layers.Dropout(0.5),
--> 53     layers.Dense(len(train_ds.class_names), activation="softmax") #_
↪Output layer
     54 ])
     55

```

AttributeError: '_MapDataset' object has no attribute 'class_names'

```
[7]: # Load dataset (before mapping)
train_ds_raw = keras.preprocessing.image_dataset_from_directory(
    dataset_path,
    validation_split=0.2,
    subset="training",
    seed=42,
    image_size=img_size,
    batch_size=batch_size
)

val_ds_raw = keras.preprocessing.image_dataset_from_directory(
    dataset_path,
    validation_split=0.2,
    subset="validation",
    seed=42,
    image_size=img_size,
    batch_size=batch_size
)

# Store class names before mapping
class_names = train_ds_raw.class_names
num_classes = len(class_names) # Number of classes

# Normalize images (apply Rescaling)
normalization_layer = layers.Rescaling(1./255)
train_ds = train_ds_raw.map(lambda x, y: (normalization_layer(x), y))
val_ds = val_ds_raw.map(lambda x, y: (normalization_layer(x), y))

# Build CNN Model
model = keras.Sequential([
    layers.Input(shape=(64, 64, 3)), # Explicit input shape
    layers.Conv2D(32, (3, 3), activation="relu"),
    layers.MaxPooling2D(),

    layers.Conv2D(64, (3, 3), activation="relu"),
    layers.MaxPooling2D(),

    layers.Conv2D(128, (3, 3), activation="relu"),
    layers.MaxPooling2D(),

    layers.Flatten(),
    layers.Dense(128, activation="relu"),
    layers.Dropout(0.5),
    layers.Dense(num_classes, activation="softmax") # Use stored num_classes
```

```

])

# Compile and train
model.compile(optimizer="adam",
              loss="sparse_categorical_crossentropy",
              metrics=["accuracy"])

model.fit(train_ds, validation_data=val_ds, epochs=50)

# Save model
model.save("cnn_emotion_model.h5")

```

```

Found 28709 files belonging to 7 classes.
Using 22968 files for training.
Found 28709 files belonging to 7 classes.
Using 5741 files for validation.
Epoch 1/50
718/718          10s 11ms/step -
accuracy: 0.2661 - loss: 1.7998 - val_accuracy: 0.3860 - val_loss: 1.5717
Epoch 2/50
718/718          6s 8ms/step -
accuracy: 0.3886 - loss: 1.5756 - val_accuracy: 0.4599 - val_loss: 1.4257
Epoch 3/50
718/718          6s 8ms/step -
accuracy: 0.4439 - loss: 1.4398 - val_accuracy: 0.4936 - val_loss: 1.3287
Epoch 4/50
718/718          6s 9ms/step -
accuracy: 0.4880 - loss: 1.3336 - val_accuracy: 0.5097 - val_loss: 1.2982
Epoch 5/50
718/718          6s 9ms/step -
accuracy: 0.5194 - loss: 1.2682 - val_accuracy: 0.5222 - val_loss: 1.2393
Epoch 6/50
718/718          6s 8ms/step -
accuracy: 0.5384 - loss: 1.2024 - val_accuracy: 0.5287 - val_loss: 1.2274
Epoch 7/50
718/718          5s 8ms/step -
accuracy: 0.5650 - loss: 1.1382 - val_accuracy: 0.5367 - val_loss: 1.2120
Epoch 8/50
718/718          6s 8ms/step -
accuracy: 0.5849 - loss: 1.0906 - val_accuracy: 0.5267 - val_loss: 1.2322
Epoch 9/50
718/718          6s 8ms/step -
accuracy: 0.5942 - loss: 1.0419 - val_accuracy: 0.5320 - val_loss: 1.2535
Epoch 10/50
718/718          6s 8ms/step -
accuracy: 0.6223 - loss: 0.9819 - val_accuracy: 0.5454 - val_loss: 1.2480
Epoch 11/50
718/718          5s 8ms/step -

```

accuracy: 0.6366 - loss: 0.9497 - val_accuracy: 0.5367 - val_loss: 1.2724
 Epoch 12/50
 718/718 6s 8ms/step -
 accuracy: 0.6536 - loss: 0.8933 - val_accuracy: 0.5438 - val_loss: 1.3048
 Epoch 13/50
 718/718 6s 8ms/step -
 accuracy: 0.6648 - loss: 0.8599 - val_accuracy: 0.5468 - val_loss: 1.3289
 Epoch 14/50
 718/718 6s 8ms/step -
 accuracy: 0.6800 - loss: 0.8250 - val_accuracy: 0.5344 - val_loss: 1.3632
 Epoch 15/50
 718/718 6s 8ms/step -
 accuracy: 0.6916 - loss: 0.7845 - val_accuracy: 0.5388 - val_loss: 1.4294
 Epoch 16/50
 718/718 6s 8ms/step -
 accuracy: 0.6997 - loss: 0.7533 - val_accuracy: 0.5431 - val_loss: 1.4466
 Epoch 17/50
 718/718 6s 8ms/step -
 accuracy: 0.7200 - loss: 0.7025 - val_accuracy: 0.5473 - val_loss: 1.5289
 Epoch 18/50
 718/718 6s 8ms/step -
 accuracy: 0.7278 - loss: 0.6900 - val_accuracy: 0.5341 - val_loss: 1.5419
 Epoch 19/50
 718/718 6s 8ms/step -
 accuracy: 0.7338 - loss: 0.6625 - val_accuracy: 0.5377 - val_loss: 1.5921
 Epoch 20/50
 718/718 6s 8ms/step -
 accuracy: 0.7338 - loss: 0.6542 - val_accuracy: 0.5382 - val_loss: 1.7300
 Epoch 21/50
 718/718 6s 8ms/step -
 accuracy: 0.7421 - loss: 0.6353 - val_accuracy: 0.5396 - val_loss: 1.7280
 Epoch 22/50
 718/718 6s 8ms/step -
 accuracy: 0.7519 - loss: 0.6211 - val_accuracy: 0.5440 - val_loss: 1.7685
 Epoch 23/50
 718/718 6s 8ms/step -
 accuracy: 0.7583 - loss: 0.5934 - val_accuracy: 0.5388 - val_loss: 1.9068
 Epoch 24/50
 718/718 6s 8ms/step -
 accuracy: 0.7700 - loss: 0.5813 - val_accuracy: 0.5419 - val_loss: 1.8704
 Epoch 25/50
 718/718 6s 9ms/step -
 accuracy: 0.7760 - loss: 0.5519 - val_accuracy: 0.5485 - val_loss: 1.8674
 Epoch 26/50
 718/718 6s 8ms/step -
 accuracy: 0.7772 - loss: 0.5433 - val_accuracy: 0.5414 - val_loss: 1.9151
 Epoch 27/50
 718/718 6s 8ms/step -

accuracy: 0.7843 - loss: 0.5257 - val_accuracy: 0.5351 - val_loss: 2.0613
 Epoch 28/50
 718/718 6s 9ms/step -
 accuracy: 0.7863 - loss: 0.5271 - val_accuracy: 0.5356 - val_loss: 2.0366
 Epoch 29/50
 718/718 6s 8ms/step -
 accuracy: 0.7908 - loss: 0.5216 - val_accuracy: 0.5388 - val_loss: 2.1731
 Epoch 30/50
 718/718 6s 8ms/step -
 accuracy: 0.7935 - loss: 0.4956 - val_accuracy: 0.5287 - val_loss: 2.2743
 Epoch 31/50
 718/718 6s 8ms/step -
 accuracy: 0.7961 - loss: 0.5014 - val_accuracy: 0.5468 - val_loss: 2.1282
 Epoch 32/50
 718/718 6s 8ms/step -
 accuracy: 0.8017 - loss: 0.4780 - val_accuracy: 0.5360 - val_loss: 2.2261
 Epoch 33/50
 718/718 6s 8ms/step -
 accuracy: 0.7961 - loss: 0.4892 - val_accuracy: 0.5414 - val_loss: 2.3269
 Epoch 34/50
 718/718 6s 8ms/step -
 accuracy: 0.8103 - loss: 0.4646 - val_accuracy: 0.5417 - val_loss: 2.2141
 Epoch 35/50
 718/718 6s 8ms/step -
 accuracy: 0.8134 - loss: 0.4580 - val_accuracy: 0.5408 - val_loss: 2.4638
 Epoch 36/50
 718/718 6s 8ms/step -
 accuracy: 0.8152 - loss: 0.4534 - val_accuracy: 0.5379 - val_loss: 2.5770
 Epoch 37/50
 718/718 6s 9ms/step -
 accuracy: 0.8144 - loss: 0.4473 - val_accuracy: 0.5396 - val_loss: 2.4033
 Epoch 38/50
 718/718 6s 8ms/step -
 accuracy: 0.8214 - loss: 0.4395 - val_accuracy: 0.5372 - val_loss: 2.5222
 Epoch 39/50
 718/718 6s 8ms/step -
 accuracy: 0.8180 - loss: 0.4428 - val_accuracy: 0.5361 - val_loss: 2.5956
 Epoch 40/50
 718/718 6s 8ms/step -
 accuracy: 0.8307 - loss: 0.4103 - val_accuracy: 0.5321 - val_loss: 2.5512
 Epoch 41/50
 718/718 6s 8ms/step -
 accuracy: 0.8296 - loss: 0.4258 - val_accuracy: 0.5354 - val_loss: 2.5477
 Epoch 42/50
 718/718 6s 8ms/step -
 accuracy: 0.8354 - loss: 0.4105 - val_accuracy: 0.5368 - val_loss: 2.6504
 Epoch 43/50
 718/718 6s 8ms/step -

```

accuracy: 0.8297 - loss: 0.4105 - val_accuracy: 0.5410 - val_loss: 2.5684
Epoch 44/50
718/718          6s 8ms/step -
accuracy: 0.8317 - loss: 0.4138 - val_accuracy: 0.5438 - val_loss: 2.7444
Epoch 45/50
718/718          6s 9ms/step -
accuracy: 0.8331 - loss: 0.4034 - val_accuracy: 0.5408 - val_loss: 2.8377
Epoch 46/50
718/718          6s 8ms/step -
accuracy: 0.8402 - loss: 0.3966 - val_accuracy: 0.5438 - val_loss: 2.7466
Epoch 47/50
718/718          6s 8ms/step -
accuracy: 0.8394 - loss: 0.3917 - val_accuracy: 0.5412 - val_loss: 2.9086
Epoch 48/50
718/718          6s 8ms/step -
accuracy: 0.8411 - loss: 0.3896 - val_accuracy: 0.5414 - val_loss: 2.9230
Epoch 49/50
718/718          6s 8ms/step -
accuracy: 0.8447 - loss: 0.3862 - val_accuracy: 0.5398 - val_loss: 2.8466
Epoch 50/50
718/718          6s 8ms/step -
accuracy: 0.8443 - loss: 0.3825 - val_accuracy: 0.5445 - val_loss: 2.9531

```

```

[9]: import tensorflow as tf
from tensorflow import keras
import numpy as np
import os
import matplotlib.pyplot as plt
import cv2
import seaborn as sns
from sklearn.metrics import confusion_matrix, classification_report

# Define paths
model_path = "/kaggle/working/cnn_emotion_model.h5" # Trained model
test_path = "/kaggle/input/fer2013/test/" # Test dataset

# Load trained model
model = keras.models.load_model(model_path)

# Define image size
img_size = (64, 64)
batch_size = 32 # Process images in batches

# Load class names from test dataset
class_names = sorted(os.listdir(test_path)) # Get emotion labels from folder
↳ names
num_classes = len(class_names)

```

```

# Function to preprocess an image
def preprocess_image(img_path):
    img = cv2.imread(img_path) # Read image
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB) # Convert to RGB
    img = cv2.resize(img, img_size) # Resize to match training input
    img = img / 255.0 # Normalize pixel values (0-1)
    return img

# Load all test images and labels
test_images, test_labels, test_file_paths = [], [], []
for class_label in class_names:
    class_folder = os.path.join(test_path, class_label)
    if os.path.isdir(class_folder):
        for file in os.listdir(class_folder):
            img_path = os.path.join(class_folder, file)
            test_file_paths.append(img_path)
            test_images.append(preprocess_image(img_path))
            test_labels.append(class_names.index(class_label)) # Convert label
↳ to index

# Convert to NumPy arrays
test_images = np.array(test_images)
test_labels = np.array(test_labels)

# Make predictions in batches
predictions = model.predict(test_images, batch_size=batch_size)
predicted_labels = np.argmax(predictions, axis=1)

# Compute confusion matrix
conf_matrix = confusion_matrix(test_labels, predicted_labels)

# Plot confusion matrix
plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt="d", cmap="Blues",
            xticklabels=class_names, yticklabels=class_names)
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("Confusion Matrix")
plt.show()

# Plot class-wise accuracy
class_accuracies = conf_matrix.diagonal() / conf_matrix.sum(axis=1)
plt.figure(figsize=(10, 5))
sns.barplot(x=class_names, y=class_accuracies, palette="coolwarm")
plt.ylabel("Accuracy")
plt.xlabel("Emotion Classes")

```

```

plt.title("Class-wise Accuracy")
plt.ylim(0, 1)
plt.xticks(rotation=45)
plt.show()

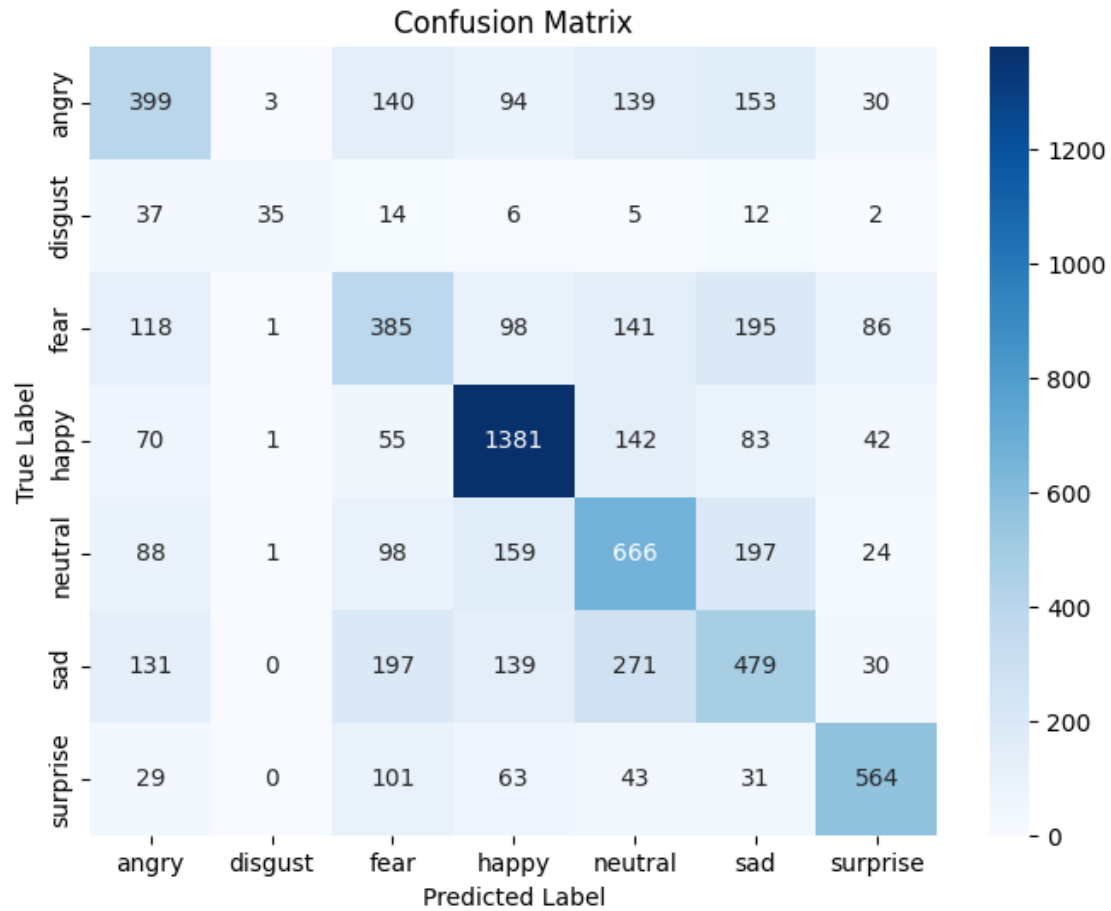
# Show images with predictions (batch-wise)
batch_size_to_show = 10 # Number of images per row
plt.figure(figsize=(12, 6))
for i in range(batch_size_to_show):
    plt.subplot(2, 5, i+1)
    plt.imshow(test_images[i])
    plt.title(f"Predicted: {class_names[predicted_labels[i]]}\nActual:␣
↪{class_names[test_labels[i]]}", fontsize=10)
    plt.axis("off")
plt.tight_layout()
plt.show()

# Print classification report
print("\nClassification Report:")
print(classification_report(test_labels, predicted_labels,␣
↪target_names=class_names))

```

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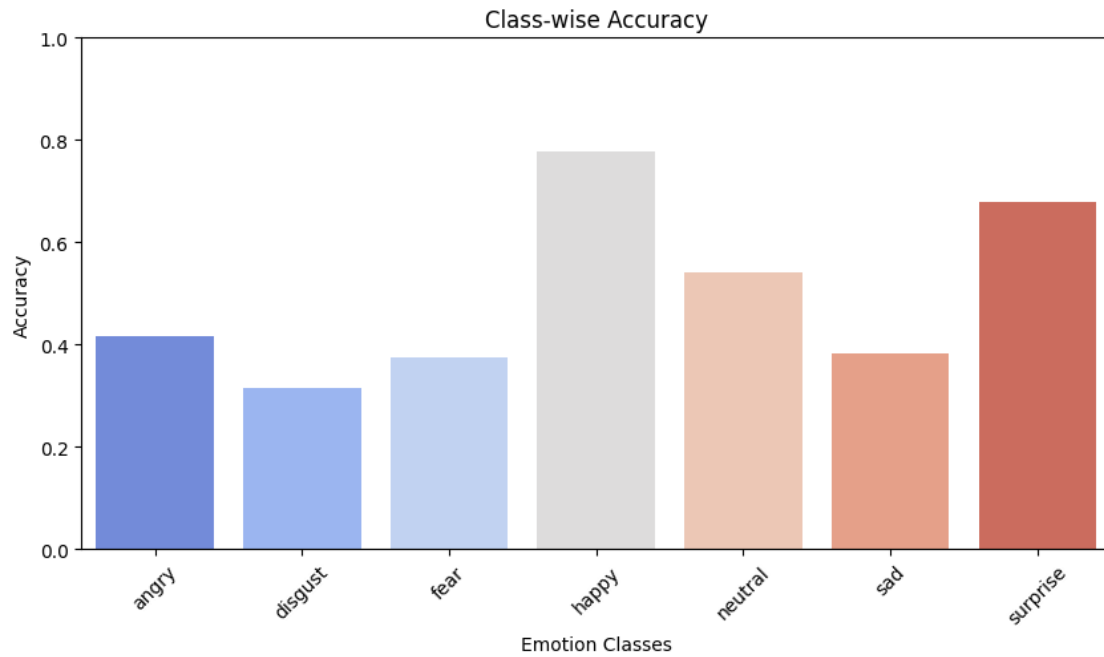
1s 3ms/step



```

/usr/local/lib/python3.10/dist-packages/seaborn/_oldcore.py:1765: FutureWarning:
unique with argument that is not not a Series, Index, ExtensionArray, or
np.ndarray is deprecated and will raise in a future version.
    order = pd.unique(vector)

```



Classification Report:

	precision	recall	f1-score	support
angry	0.46	0.42	0.44	958
disgust	0.85	0.32	0.46	111
fear	0.39	0.38	0.38	1024
happy	0.71	0.78	0.74	1774

neutral	0.47	0.54	0.50	1233
sad	0.42	0.38	0.40	1247
surprise	0.72	0.68	0.70	831
accuracy			0.54	7178
macro avg	0.58	0.50	0.52	7178
weighted avg	0.54	0.54	0.54	7178